Errata to the stored explanations, 2026-11-11

Scope. In verifier-benchmark run 1, blind verifiers found genuine errors in several original explanations (`experiments/verifier_bench/runs/run1/ADJUDICATION.md`). Each of those errors was replicated by more than one verifier. I then ran a sweep with `explaneat verify recheck` (v2, including the free-text checks) over every stored explanation: the 60 10-seed records, the 3 AAAI paper genomes, the 13 species-lead records and the 4 complex-real records (80 in all). The sweep produced 53 mismatch findings and 35 rounding, uncited-citation, lint or formula warnings (`sweep_triage.csv`). Every candidate was checked against the model with `verify probe`, `subgroup`, `ablate` or `linearise`, or recomputed from node weights and values. A value was corrected only once it was confirmed. Most sweep mismatches turned out to be the tool misreading the text, for example:

- a conditional share ("46% of non-voicemail rows") compared with the overall share;
- an AUC read as a readout weight;
- a formula written in raw units.

They are marked "not an error" in the CSV, each with the check that settled it.

Before any change, the records, the plain-language layers and the database `Explanation.operations` of every affected explanation were snapshotted to `snapshot/`, with sha256 values in `SHA256SUMS` and `MANIFEST.json`. The git tag is `explanations-pre-errata-2026-11-11`.

Errors

"Blind" = found by the blind verifiers (ADJUDICATION.md). "Sweep" = flagged by `recheck` v2. "Confirm" = found while confirming a neighbouring error. "Layer" = `experiments/<study>/plain_language/<label>.json`; "DB" = `Explanation.operations`; every DB hypothesis edit is mirrored in the layer's copy (`parts[].original_hypothesis`). All errors below were **corrected**.

#	Explanation	Location	Said	Correct	Confirmed by	Found by
E1	heart-s5	DB `ischaemia_pathway` hypothesis and `evidence[1].narrative`	35 low-BP patients (290 on, 204 off): mean predicted risk 0.13	0.18 (0.1813)	node values on train: 135 with 290 on, 35 of them with 204_a/204_b off, mean risk 0.1813	Blind
E2	adult-s5	DB `age_profile` hypothesis	Thai respondents "(27 rows)"	26 rows	train rows with native-country = Thailand: 26 (all 26 have 2134 on)	Blind
E3	adult-s5	DB `age_profile` hypothesis	`2990=relu(1.446 age-0.48)`	`2990=relu(1.446 age+0.137*2134-0.48)` (the Thailand unit 2134 also feeds 2990)	incoming edges of 2990: age 1.446, 2134 0.1374, bias -0.481; `recheck` mismatch 4.45 on the 26 Thai rows, 0 after	Sweep (mismatch)
E4	adult-s5	layer `blocks[3][0]`, `readdown.ann_125.evidence[0]`, `summary[11]` (claims)	"about 50 hours a week" cited C143 only	cites C142 (ann_125's 50-hour threshold claim) as well; "(no further effect above that)" added, the shape phrase the lint requires for C142	`recheck` prose_uncited; C142 threshold 49.9 h	Blind, Sweep (warn)
E5	churn100-s4	DB `area415_state47` hypothesis	595 "zero in area 415"; "Zero in area 415 (2,016 rows)"	zero for 1,986 of the 2,016 area-415 rows; 1.03 (logit +0.21) for the 30 Washington (state 47) customers in area 415	node 595 by area code × state 47: area 415 non-WA 0 (n = 1,986), area 415 WA 1.0334 (n = 30)	Blind
E6	churn100-s4	same hypothesis	offset "˜0.15" / "0.16 for other areas (logit +0.03)"	0.11 (0.1058), logit +0.02	same table: areas 408 and 510 without WA, 595 = 0.1058; × 0.203 = 0.021	Confirm
E7	heart-s2	layer `blocks[1][2]`, `readdown.ann_31.evidence[1]`, `summary[7]`	"but for patients with a reversible defect on the thallium scan it rises in a jump: for a typical patient from about 12% at 42 to about 28% at 66 years"	the 12% to 28% rise is for the typical patient (normal scan); "but for a typical patient with a reversible defect on the thallium scan this part is off at every age"; now also cites C33	`probe --sweep age=35:75:9 --sweep thal=3,7`: ann_31 contribution 0 at every age with thal = 7; `subgroup "thal == '7'"`: the part acts on 3 of 90	Blind
E8	churn100-s1	DB `churnModelRoot` hypothesis; record `summary`	voice-mail plan "-2.37" logit for holders	-1.96	direct weight -0.8641 / sd 0.4415 = -1.957; `probe --sweep "voice mail plan=0,1" --wide`: score -3.7651 → -5.7225	Blind
E9	churn100-s1	same hypothesis; record `summary`	international plan "+2.67" logit for holders	+2.62	0.769 / sd 0.2932 = 2.623; probe score change 2.633 less ann_81's +0.011	Confirm
E10	churn100-s1	layer `blocks[2][2]`, `readdown.ann_79.evidence[2]`, `readdown.ann_83.evidence[2]`, `summary[11]`	"about 5%"	about 4% (cited value 4.45%, rounded twice)	`recheck` prose_rounding	Blind, Sweep (warn)

#	Explanation	Location	Said	Correct	Confirmed by	Found by
E11	monk2-s0	DB root `upper_edge_count_classifier` `evidence[2].payload` (a1=3); record `resisted`	"108/108 pos"	107/107	`subgroup "a1 == '3'"`: n = 107, all predicted yes	Blind
E12	churn100-s7	layer `blocks[0][2]`, `readdown.ann_80.evidence[2]` (and summary copy)	"about 15%"	about 14% (14.48%)	lint; `recheck` prose_rounding	Lint, Sweep (warn)
E13	heart100-s0	layer `lead[0]` (and summary copy)	"about 20%"	about 19% (19.3%)	lint	Lint, Sweep (warn)
E14	heart-s5-c1	layer `blocks[0][0]` (and summary copy), `readdown.ann_35.evidence[3]`	"about 28% at 118"; "about 27% at 66 years"	about 27% (27.50%); about 26% (26.45%)	`recheck` prose_rounding	Sweep (warn)
E15	heart100-s7	layer `blocks[2][1]`, `readdown.ann_38.evidence[2]` (and summary copy) (claims)	"about 168 beats per minute" cited C82 only	cites C83 (ann_38's 168 bpm threshold) as well, with its spread phrase "(between 160 and 176 beats per minute for different patients)"	`recheck` prose_uncited; C83 threshold 168.2	Sweep (warn)

Listed, not corrected:

- **AAAI genomes** (materials not to be changed):
  - test-split AUCs that `recheck` cannot check on train (heart-paper ×3, adult-paper);
  - activity shares written as 70% and 17.5%, recomputed as 68.2% and 19.0% (heart-paper);
  - closed-form display warnings (node ids in the written LaTeX);
  - a monk2-paper direction false positive.
- **Approximate formulas** (`freetext_formula` warn; each is stated as "nearly" or "simplified", or is off on a few rows only): heart-s2, heart-s2-c1, churn200-s0, heart100-s4, monk2-s4.
- **Closed-form display warnings:** heart-s0, heart-s6.
- **heart100-s6.** "Block contributes train AUC 0.905 -¿ 0.958". Removing both readouts of the block gives 0.9036. The method is not stated, so this is ambiguous.
- **c4-tuned-s0.** "29 squares" is correct (ann_44 reads 29 inputs, with one direction claim each). The number happens to match the uncited C146.
- **heart-s2, generated text.** The source of E7 is claim C30's computed shape, which lists "step" as an exception for reversible-defect patients. That exception holds only for the 29 of 90 such patients whose part is on at a young age. The auto-generated `template` text ("moves mostly in a jump") still carries it. Claims and templates are code output, so they were not hand-edited.
- **Stale rendered copies.** The verification packets in `experiments/10seed/verification/` (`02_heart-s5.md`, `16_monk2-s0.md`, `20_heart-s2.md`, `23_adult-s5.md`, `packets.html`, `live_checks.json`, `supplementary/C100-01_churn100-s4.md`, `C100-02_churn100-s1.md`) quote the old text. They were not regenerated. The verifier-benchmark items are frozen copies and were left as they are.

Effect on reported numbers

There is none. Every error is in explanation text: hypotheses, evidence narratives or payloads, record summary/resisted, and the plain-language drafts. The changes are:

- **Structure:** no structural operation changed. Before writing, the correction script checked that the subgraphs, entry and exit nodes and children of every operation were identical.
- **Recheck:** `check_explanation` and `get_coverage` were re-run on all ten corrected genomes (`check_after.json`). Structural and compositional coverage (1.0), hierarchy completeness, function preservation (max error $\leq$ 1e-15), closed-form error and operation counts all equal the stored `verified_by_orchestrator` values.
- **CSVs:** `explanations.csv` was rebuilt. Only monk2-s0's `resisted` text changed; the heart100 and churn CSVs are byte-identical.
- **Unchanged files:** `summary.md`, `ch7_table.tex`, the runs CSVs and the AAAI materials.
- **`mike_verification`:** untouched. Each corrected record carries an `errata` list (date, location, was, now, ref to this file).

After the correction (`after.json`; the edits themselves are in `apply_errata.py`), `recheck` reports no remaining genuine mismatch on the ten corrected explanations. Its leftover findings are the false positives above (churn100-s7, heart-s5-c1, heart100-s7) and the heart-s2 approximate-formula warning. All corrected drafts pass `plain-language lint`. These outputs were rebuilt:

- the nine per-genome PDFs (`~/.explaneat/exports/plain-language/<label>.pdf`);
- the first-10 verification pack (heart-s5 is in it), whose checklist pages now print "Errata applied 2026-11-11";
- heart-s5-c1's certificate (`experiments/species_leads/verification/heart-s5-c1/heart-s5-c1.pdf`).

The species-leads combined `review.pdf` files were built with earlier export code. Rebuilding them would change every lead's pages, so they were left alone. They still show heart-s5-c1's two old percentages until `experiments/species_leads/make_pack.py` is re-run.

Update (2026-11-11, commit `4d619be`): the verification packets in `experiments/10seed/verification/`

and the species-leads verification pack, including its combined `review.pdf`, have since been regenerated. No rendered copy now shows the old text.